

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

**COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1105**

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing techniques. (BL4,BL5)
Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:40

Annexure A

MOCK INTERVIEW

Code of Conduct:

(192321357)

I Rashmi Naware (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rashmi Naware

Annexure A

Mock Interview – Sample Questions (5 Questions)

Mock Interview Test Format for Object Oriented Analysis and Design (OOAD)

Section A	Self Introduction & Fundamentals	3 Minutes	5
Section B	Technical Questions (OOAD Concepts)	7 Minutes	10
Section C	UML Modeling & Design	7 Minutes	10
Section D	Case Study / Problem Solving	8 Minutes	10
Section E	HR & Career-Oriented Questions	5 Minutes	5

SECTION A – SELF INTRODUCTION (5 Marks)

Evaluation Criteria

- Communication Skills
- Confidence
- Technical Awareness
- Academic Background

Sample Questions

1. Tell me about yourself.
 2. Why did you choose Computer Science?
 3. Explain your favorite subject in OOAD.
 4. What software development tools have you used?
 5. Describe a project where you applied object-oriented concepts.
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SECTION B – TECHNICAL QUESTIONS (10 Marks)

Basic Level

Q1. What is Object-Oriented Analysis and Design?

Q2. Differentiate between Analysis and Design.

Q3. Explain Encapsulation with an example.

Q4. What is Inheritance? Mention its types.

Q5. Define Polymorphism and its benefits.

Q6. What is Abstraction?

Q7. Difference between Aggregation and Composition.

Q8. What is Coupling and Cohesion?

Q9. What is a Class and an Object?

Q10. Explain Object Responsibility.

SECTION C – UML MODELING (10 Marks)

Q11. What is UML?

Q12. When would you use a Use Case Diagram?

Q13. Draw and explain a Class Diagram for a Library Management System.

Q14. Differentiate Sequence Diagram and Activity Diagram.

Q15. Explain State Diagram with an example.

Q16. What is a Deployment Diagram?

Q17. How do Component Diagrams help software architects?

Q18. What is Multiplicity in UML?

SECTION D – CASE STUDY / PROBLEM SOLVING (10 Marks)

Case Study 1

A hospital wants a system to manage:

- Patients
- Doctors
- Appointments
- Billing

Questions

1. Identify the classes.
 2. Draw a simple class diagram.
 3. Suggest relationships among classes.
 4. Which UML diagrams would you use for analysis?
 5. Explain how inheritance can be applied.
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Case Study 2

An Online Shopping System supports:

- Customers
- Products
- Cart
- Orders
- Payments

Questions

1. Identify actors and use cases.
 2. Design a use case diagram.
 3. Suggest suitable design patterns.
 4. Explain how polymorphism can be used in payment processing.
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SECTION E – HR & PROFESSIONAL QUESTIONS (5 Marks)

Q19.Why should we hire you?

Q20.What are your strengths and weaknesses?

Q21.How do you handle project deadlines?

Q22.Have you worked in a team project? Explain your role.

Q23.Where do you see yourself in five years?

RUBRICS FOR CALCULATION:

MOCK INTERVIEW

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Section B - Technical Questions.

Q1. What is Object-Oriented and Design?

Ans Object Oriented Analysis and Design is a software engineering approach that models a system as a collection of interacting objects.

2. Differentiate Between Analysis and Design.

Ans Analysis focuses on understanding the problem and requirements while Design focusses on deciding how the system will be implemented.

3. Explain encapsulation with an example.

Ans Encapsulation combines data and methods inside a class and controls access to the data.

4. What is inheritance? Mention its types.

Ans Inheritance allows a new class to acquire properties and methods from an existing class. Types include single, multilevel, hierarchical, multiple and hybrid inheritance.

5. Define Polymorphism and its benefits.

Ans Polymorphism means one interface can have different implementations depending on the object. It improves flexibility and reusability.

6. What is Abstraction?

Ans Abstraction hides unnecessary implementation details and shows only the essential features of an object.

7. Difference between Aggregation and Composition.

Ans Aggregation represents a weak whole-part relationship where parts can exist independently, whereas Composition represents a strong relationship.

8. What is coupling and cohesion?

Ans Coupling is the degree of dependency between different classes or modules.
Cohesion is how closely related the responsibilities within a module are.

9. What is a class and an object?

Ans A class is a blueprint that defines the properties and behaviours of objects. An object is a real instance of a class.

10. Explain Object Responsibility.

Ans Object Responsibility refers to the tasks or actions that an object is responsible for performing.

Section C - UML MODELING

11. What is UML?

Ans Unified Modeling Language is a standard visual language used to model and document software systems.

12. When would you use a Use Case Diagram?

Ans A Use Case Diagram is used during requirement analysis to show the system's actors and their interactions with the system.

13. Draw and explain a class diagram for Library Management System.

Ans A Library Class Diagram can contain classes such as Book, member, Librarian and Transaction with their relationships.

14. Differentiate Sequence Diagram and Activity Diagram.

Ans A Sequence Diagram shows interactions between objects in a particular time sequence. An activity diagram shows the workflow of activities performed.

15. Explain state Diagram.

Ans A State diagram represents the different states of an object and the events that cause it to change state.

16. What is a Deployment Diagram?

Ans. A deployment Diagram shows the physical arrangement of hardware and software components in a system.

17. How do components Diagram help software architect?

Ans. Component Diagrams show the major software components and the dependencies between them.

18. What is Multiplicity in UML?

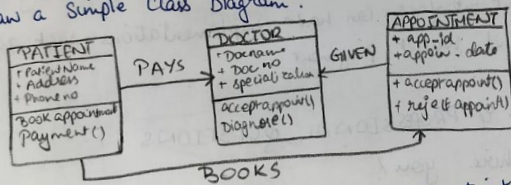
Ans. Multiplicity specifies how many objects of one class can be associated with an object of another class.

Section D - Case Study.

1. Hospital Management System.

1. Identify the classes → Main Classes are Patient, Doctor, Appointment & Billing.

2. Draw a Simple Class Diagram.



3. Suggest Relationship among Classes → A patient has appointments with doctors and an appointment connects patient and doctor.

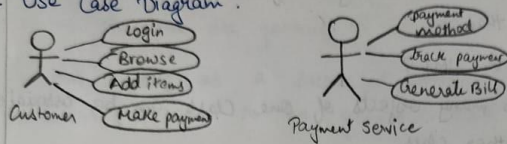
4. Which UML diagram would you use for analysis → A use case diagram can identify users and system requirements during analysis.

5. How inheritance can be applied → A general person can be created with common details such as name and contact information. 'Patient' and 'doctor' class can inherit these properties from the person class.

3) Online Shopping System.

1. Actors and Use Cases → The main actor is the customer who can browse products, add products to cart, place orders and make payments. Other Actors are Payment Service and Administrator.

2. Use Case Diagram.



3. Suggest suitable design patterns → The factory pattern can create different product or payment objects while strategy pattern can handle different payment methods.

4. Explain how polymorphism can be used in payment processing → A common payment interface can have implementations such as Credit Card, UPI and Wallet payment.

SECTION E - HR & PROFESSIONAL QUESTIONS

19. Why should we hire you?

→ You should hire me because I am willing to learn, adapt quickly and apply my technical knowledge to practical problems.

20. What are your strengths and weaknesses?

→ My strengths are quick learning, problem solving and persistence. My weakness is that sometimes I am a bit judgy and over-confident.

21. How do you handle project deadlines?

→ I divide the project into smaller tasks, prioritize important work and set deadlines for each task.

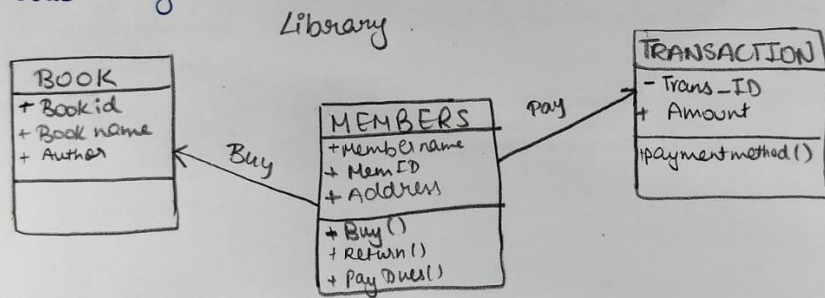
22. Have you worked in a team project? Explain Your role.

→ Yes, I have worked in various team projects. The role I took on was that of a team leader. I guided my fellow teammates into successfully completing the projects that we were assigned with.

23. Where do you see yourself in 5 years?

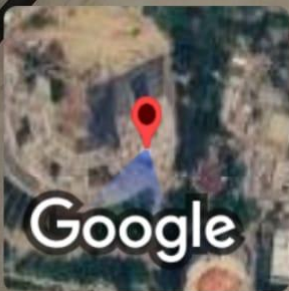
→ I see myself as a successful entrepreneur running a profitable business, I want to be generous and kind and knowledgeable.

13. Class Diagram.





GPS Map Camera



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